CODE & COMPLIANCE 2026 · BLOCK 4 · BRUSSELS

Physical machinery documentation turned into compliance assets

The EnduraDocs case and iiRDS implementation

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I’ll show what we built it for machinery and where it still fails.

EnduraDocs GmbH

- Founded 2025 in Aachen
- Digital machinery documentation as a Service
- iiRDS - intelligent information Request and Delivery Standard
    - industry standard by tekom as the foundation
- Member of the iiRDS Consortium
- Forthcoming published Request API as open source

THE STARTING POINT

One machine, five EU laws

A typical machine today

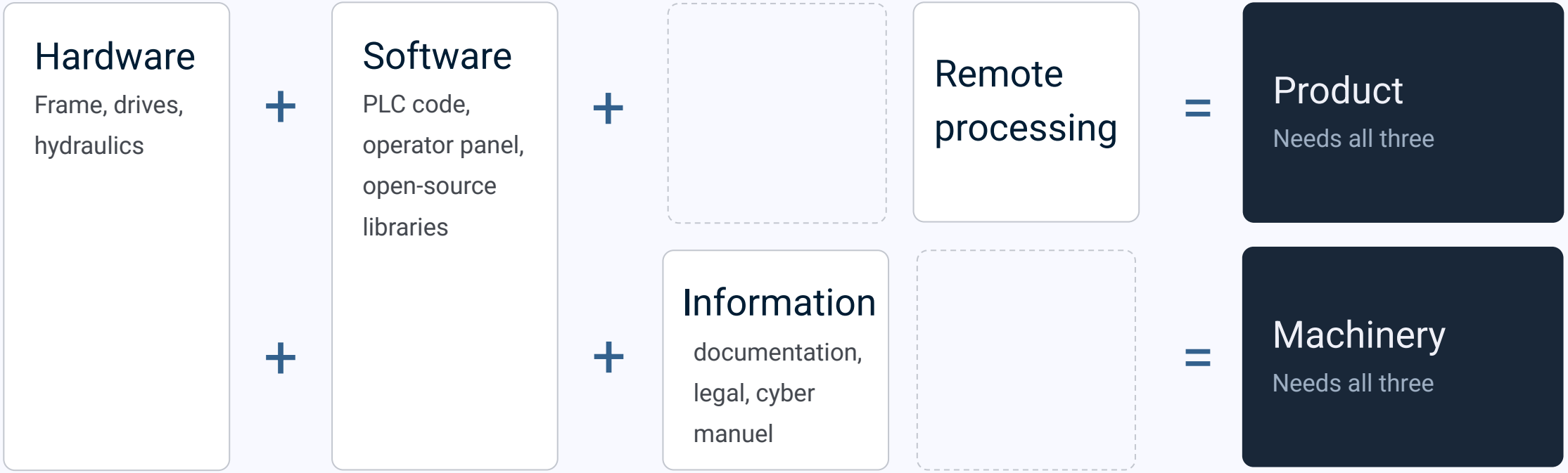
Has software inside

Connects to a network

Ships with instructions

Law	Applies from
Data Act, access by design	12 Sep 2026
Product Liability Directive	09 Dec 2026
Machinery Regulation	20 Jan 2027
Battery passport, if it has an industrial or EV battery	18 Feb 2027
Cyber Resilience Act (reporting since Sep 2026)	11 Dec 2027

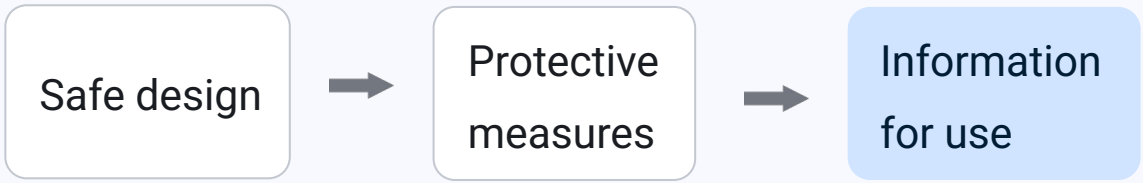
A product is hardware, software and information



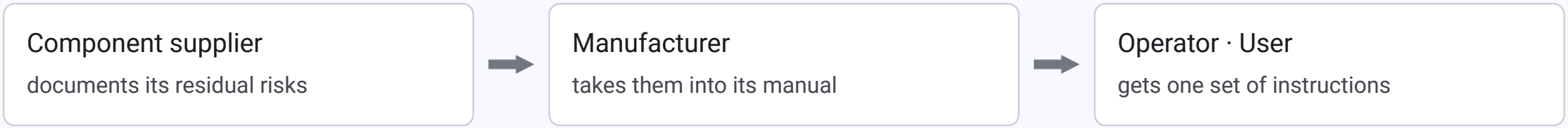
If one of the three is missing, the product is not complete.

An error in the information is an error in the product.

Residual risk travels down the chain



INSIDE MANUFACTURER · EN ISO 12100



ALONG THE SUPPLY CHAIN

CRA Recital 55: risks that design cannot remove can be addressed by informing users.

Software Components clear duties, but no common format

Commercial supplier

Proprietary or monetised open source. A CRA manufacturer with its own user information

Liable as component maker (PLD Art. 8(1)(b))

Open-source software steward

Supports open source meant for commercial use

Light regime only (CRA Art. 24)

In both cases

Information comes in each supplier’s own format

Owner and version get lost on the way into the manufacturer’s manual

In every case the information arrives in the supplier’s own format, and owner and version get lost. We need a common format that travels with the component.

What turns documentation into a compliance asset

Origin

Every part names who is responsible for it, component supplier or manufacturer.

Because supplier information loses its owner on the way

iiRDS

Versioned

Correct for the product version in use, and updated with every relevant fix.

Because the CRA asks for years, not one day

iiRDS + IPFS

Durable

Available and checkable for 10+ years, without depending on one vendor.

Because vendors may not last ten years

IPFS

iiRDS: a standard that says what each document is

product.iirds (a ZIP file)

metadata.rdf
What each file is, for what, for whom

Content
PDF, HTML, video, in any format

Supplier packages
Nested, each with its own metadata

The metadata

- product variant and component
- information type: operating, maintenance, security
- lifecycle phase and audience
- language and revision
- the responsible party

Open standard (tekom, iiRDS Consortium). Any tool writes it, any tool reads it.

iiRDS makes responsibility machine-readable

What MR and CRA ask for		How iiRDS expresses it
Manufacturer and contact point	CRA Annex II 1–2	A party with a role and contact details
Product identification	MR 10(5), CRA 13(15)	Product, variant, component, serial or type
Security manual, update instructions	Annex II 8	Typed documents next to the operating instructions
Information for integrators	Annex II 8f	Supplier packages nested in the manufacturer’s
Correct over the support period	Annex I 2(20)	Every revision linked to the same document

The supplier’s information now travels with the component, owner and version intact.

IPFS: data addressed by their content

TODAY: BY LOCATION

`https://acme.com/SD-34/manual.pdf`

Works while that server exists.
The file can change without anyone noticing.

IPFS: BY CONTENT

`ipfs://bafy…q4e (the CID)`

The address is a hash of the data. Same bytes, same address, anywhere. Any change gives a new address.

Pinning

Any node can keep and serve a copy

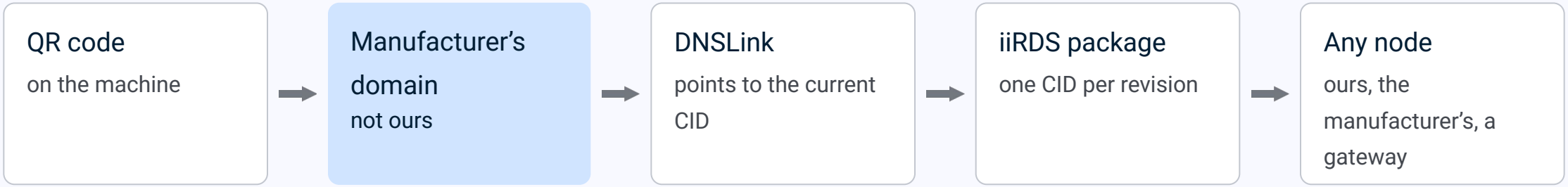
CAR file

Moves content offline, for example on USB

DNSLink

A stable domain name that points to the current CID

IPFS allows to pin data without EnduraDocs involvement



Update

New revision, new CID, the pointer moves. Old revisions stay available (CRA Art. 13(18)).

Exit

Pin the CIDs elsewhere, repoint DNS. Works if we or the manufacturer stop (CRA Art. 13(23)).

Offline plant

CAR file on USB, checked against the CID on arrival. Fits MR Art. 10(7)(b).

Clear separation of duty and care within the ecosystem

Party	Owns
Manufacturer	Writes the manual, is named in it, owns the domain
Component supplier	Delivers its own package, nested in the manufacturer’s
Operator	Scans the QR code, can keep its own copy
Anyone in the chain	Can serve a copy; the CID proves it is unchanged
EnduraDocs GmbH	Hosting, pinning, the API. Replaceable.

With an owner, a version history and no dependency on us, documentation becomes a compliance asset.

WHERE IT BREAKS

Four things that do not work yet

We have not solved these. Help welcome.

1 **Confidential content**
Public IPFS is readable by anyone with the CID. Per-file encryption is in progress, with known gaps.

2 **A CID is not yet evidence**
It proves the bytes are unchanged, not when or by whom they were published.

3 **Where the SBOM goes**
Inside the package, linked from it, or kept apart. Undecided.

4 **The standard is hard to use**
Users struggle most with defining the metadata. Tooling is the bottleneck.

Machinery documentation becomes a compliance asset when it has an owner, a version history and outlives its vendor.

The same data serves the Machinery Regulation, the CRA, liability evidence and later the product passport. So the compliance work pays off more than once.

WHAT WE ARE LOOKING FOR

People to build this as a shared, foundation-hosted open-source project · Review of the CRA Annex II mapping · Pointers on SBOM placement and evidence

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